

6. (a) Explain and prove Schroeder Bernstein theorem. (10)
 (b) What is a Perfect Graph? Explain with example. (5)
7. (a) What is Integral domain? Explain with example. (5)
 (b) Consider an algebraic system $(Q, *)$ where Q is a set of rational numbers and $*$ is binary operation defined by:

$$a*b = a + b - ab$$

Determine whether $(Q, *)$ is a group. (10)

Roll No.

Total Pages : 4

301401

May 2019

B.Tech. (CE/IT/CSE)- IVth Semester

DISCRETE MATHEMATICS

(PCC-CS-401)

Time : 3 Hours]

[Max. Marks : 75

Instructions :

- (i) It is compulsory to answer all the questions (1.5 marks each) of Part-A in short.
 (ii) Answer any four questions from Part-B in detail.
 (iii) Different sub-parts of a question are to be attempted adjacent to each other.

PART-A

1. (a) For any set A and B , show that
 $(A \cap B) \cup (A - B) = A$. (1.5)
 (b) Define equivalence relation with example. (1.5)
 (c) How many 11 letter words can be formed using letters from the word "INSTITUTION" ? (1.5)
 (d) Show that $1^2 + 2^2 + 3^2 + \dots + n^2 = n(n+1)/6$. (1.5)

- (e) What is a bipartite Graph? (1.5)
- (f) Define Euler Formula. (1.5)
- (g) What is a cut point and bridge in graphs ? (1.5)
- (h) What are quantifiers in propositions? (1.5)
- (i) What are bijective functions? (1.5)
- (j) List the applications of trees. (1.5)

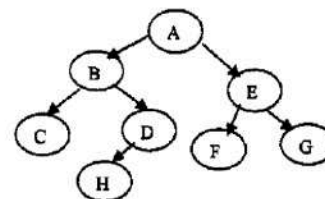
PART-B

2. (a) Let f be a function from the set of integers such that $f(x) = x + 1$. Is f invertible, and if it is what is its inverse? (5)
- (b) Seven women and nine men are on faculty in the mathematics department in a university. In how many ways a committee of five members of the department can be constructed if at least one woman and atleast one man must be in the committee?
- (c) Define Cantor's Diagonal Argument. (5)
3. (a) Show that following implication is tautology :
 - (i) $(p \Rightarrow q) \vee r \Leftrightarrow [(p \vee r) \Rightarrow (q \vee r)] \Rightarrow$
 - (ii) $(p \wedge q \Rightarrow r) \Leftrightarrow (p \Rightarrow r) \vee (q \Rightarrow r)$ (10)
- (b) Find the validity of the following Argument :
 Either Ram is a good boy or Rahul is a good boy.
 Ram is not a good boy therefore Rahul is a good boy. (5)

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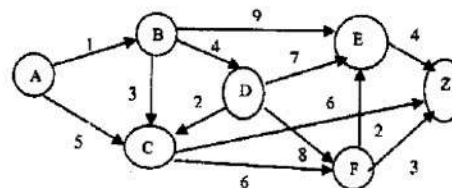
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4. (a) Determine the Inorder, preorder and postorder Traversal of following tree : (6)



- (b) Write short notes on following :
 - (i) CNF.
 - (ii) Hamiltonian Graph.
 - (iii) Subgroups in algebraic structures. (9)

5. (a) Find the shortest distance between A and Z : (10)



- (b) What is a chromatic number in a graph?. Determine the chromatic number for the following graph : (5)

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[P.T.O.]